Atmospheric Propagation

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1 Introduction

These notes are concern fully correlated atmospheric propagation. We start with the stochastic Helmholtz equation, introduce important length scales and nondimensionalize.

2 Helmholtz Equation for Atmospheric Propagation

A monochromatic EM source $E(\tilde{x}, \tilde{y}, \tilde{z}) = A(\tilde{x}, \tilde{y}, \tilde{z}) \exp\bigl(-i\omega\tilde{t}\bigr) + c.c.$, propagating in a nondispersive static atmosphere follows a stochastic Helmholtz equation [Anderson Phillips, "Laser Beam Propagation through random media"]

$$\begin{aligned} &\nabla^2 \tilde{A} + k^2 (n_0 + n(\tilde{x}, \tilde{y}, \tilde{z}))^2 \tilde{A} = 0 \quad \text{for} \quad -\infty < \tilde{x}, \tilde{y} < \infty, \quad 0 < \tilde{z} < L, \\ &\text{where} \quad \tilde{A}(\tilde{x}, \tilde{y}, 0) = \tilde{A}_i(\tilde{x}, \tilde{y}) \exp(i\Phi(\tilde{x}, \tilde{y})), \end{aligned} \tag{1}$$

and $\tilde{A}_i(\tilde{x}, \tilde{y})$ is of finite support, i.e., $\tilde{A}_i(\tilde{x}, \tilde{y}) = 0$ when $\sqrt{\tilde{x}^2 + \tilde{y}^2} > D_i$. Note, $k = \frac{\omega}{c} = \frac{2\pi}{\lambda}$.

2.1 Atmospheric Statistics

This section contains the most important part of these notes. Normally, the stochastic part of the atmosphere is described by the **structure function**,

$$\begin{aligned} \mathbb{E}\bigl[(n(\tilde{\boldsymbol{r}}_1) - n(\tilde{\boldsymbol{r}}_2))^2\bigr] &= C_n^2 \begin{cases} l_0^{-4/3} \|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\|^2 & 0 < \|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\| \le l_0 \\ \|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\|^{2/3} & l_0 \le \|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\| \le L_0 \end{cases} \\ &= \mathbb{E}\bigl[n^2(\tilde{\boldsymbol{r}}_1)\bigr] + \mathbb{E}\bigl[n^2(\tilde{\boldsymbol{r}}_2)\bigr] - 2\mathbb{E}[n(\tilde{\boldsymbol{r}}_1) n(\tilde{\boldsymbol{r}}_2)] \end{aligned} \tag{2}$$

Note, this is the only statistic provided to us and this is the only theoretical information we have to act on. It is common to make the assumption that the index of refraction variations $n(\tilde{\boldsymbol{r}})$ follow a **stationary** or **homogeneous** process [Wiki], at least in the wide-sense [1], which imposes that the $n(\tilde{\boldsymbol{r}})$ has constant mean and an auto-covariance that only depends on the difference in locations:

$$\mathbb{E}[n(\tilde{\boldsymbol{r}}_1)] = \mu_n, \quad \mathbb{E}[n(\tilde{\boldsymbol{r}}_1) n(\tilde{\boldsymbol{r}}_2)] = C_{cos}(\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2) : \mathbb{R}^3 \to \mathbb{R} \tag{3}$$

In addition, it is commonly assumed that $n(\tilde{\boldsymbol{r}})$ is mean zero and **isotropic**, implying that $F(\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2) = F(\|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\|)$. This implies that $\mathbb{E}\big[n^2(\tilde{\boldsymbol{r}}_1)\big] = \mathbb{E}\big[n^2(\tilde{\boldsymbol{r}}_2)\big] = C_{cov}(\|\boldsymbol{0}\|) = \sigma_n^2$ and gives

$$\begin{aligned}\mathbb{E}\big[(n(\tilde{\boldsymbol{r}}_1) - n(\tilde{\boldsymbol{r}}_2))^2\big] &= C_n^2 \begin{cases} l_0^{-4/3}\|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\|^2 & 0 < \|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\| \le l_0 \\ \|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\|^{2/3} & l_0 \le \|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\| \le L_0 \end{cases} \\ &= \mathbb{E}\big[n^2(\tilde{\boldsymbol{r}}_1)\big] + \mathbb{E}\big[n^2(\tilde{\boldsymbol{r}}_2)\big] - 2\mathbb{E}[n(\tilde{\boldsymbol{r}}_1)n(\tilde{\boldsymbol{r}}_2)] \\ &= 2\sigma_n^2(1 - C_{cor}(\|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\|))\end{aligned} \tag{4}$$

where we let $C_{cor}(\cdot) = C_{cov}(\cdot)/\sigma_n^2$.

so

$$C_{cor}(\|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\|) = 1 - \frac{C_n^2}{2\sigma_n^2}\begin{cases} l_0^{-4/3}\|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\|^2 & 0 < \|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\| \le l_0 \\ \|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\|^{2/3} & l_0 \le \|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\| \le L_0 \end{cases} \tag{5}$$

If we assume that $|C_{cor}(\|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\|)| \ll 1$ for $\|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\| \gg L_0$, then letting $\|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\| = C_0 L_0$ for $C_0 > 1$ (C_0 is just an adjustment factor) we can write

$$0 \approx 1 - C_0^{2/3}\frac{C_n^2 L_0^{2/3}}{2\sigma_n^2}, \tag{6}$$

or

$$\sigma_n^2 \approx C_0^{2/3}\frac{C_n^2 L_0^{2/3}}{2} \implies \frac{C_n^2}{2\sigma_n^2} = \frac{2C_n^2}{2C_0^{2/3}C_n^2 L_0^{2/3}} = \frac{1}{C_0^{2/3}L_0^{2/3}} \tag{7}$$

and

$$C_{cor}(\|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\|) = 1 - \frac{1}{C_0^{2/3}}\begin{cases} \frac{1}{(L_0 l_0^2)^{2/3}}\|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\|^2 & 0 < \|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\| \le l_0 \\ \frac{1}{L_0^{2/3}}\|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\|^{2/3} & l_0 \le \|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\| \le L_0 \end{cases} \tag{8}$$

2.1.1 Atmospheric Scales

Quantity	**Symbol**	**Characteristic Value(s)**	**Ref.**
Wavelength	λ	10^{-6} (meters)	[]
Wavelength	$k = \frac{2\pi}{\lambda}$	10^{6} (meters^{-1})	
Kolmogorov Inner-scale	l_0	$10^{-3} - 10^{-2}$ (meters)	Wheelon, 2005 ,page 32
Kolmogorov Outer-scale	L_0	$10 - 10^{3}$ (meters)	Wheelon, 2005 ,page 32
Index Structure Constant	C_n^2	$10^{-20} - 10^{-12}$ (meters$^{-2/3}$)	Wheelon, 2005 ,page 32
Index Variance	σ_n^2	$C_0^{2/3}(10^{-18} - 10^{-10})$	–
Aperture Diameter	D_{ap}	$10^{-2} - 10^{0}$ (meters)	
Propagation Distance	L	$10^{3} - 10^{4}$ (meters)	

3 Nondimensionalization

Quantity	Scale	New Variable
$\tilde{z}$	l_z	$\tilde{z} = l_z z$
$\tilde{x}$	$l_{x,y}$	$\tilde{x} = l_{x,y} x$
$\tilde{y}$	$l_{x,y}$	$\tilde{y} = l_{x,y} y$
$n(\tilde{x}, \tilde{y}, \tilde{z})$	σ_n	$n(\tilde{x}, \tilde{y}, \tilde{z}) = \sigma_n n(x, y, z)$
$\tilde{A}(\tilde{x}, \tilde{y}, \tilde{z})$	A_0	$\tilde{A}(l_{x,y}\tilde{x}, l_{x,y}\tilde{y}, l_z\tilde{z}) = A_0 A(x, y, z)$

$$\begin{gathered}\frac{1}{l_z^2}\frac{\partial^2 A}{\partial z^2} + \frac{1}{l_{x,y}^2}\nabla_\perp^2 A + k^2 n_0^2 A + 2k^2 n_0 \sigma_n n(x,y,z)A + k^2\sigma_n^2 n^2(x,y,z)A = 0 \\ \text{for} \quad -\infty < x, y < \infty, \quad 0 < z < \frac{L}{L_0} \quad \text{where} \quad A(x,y,0) = A_i(x,y)\exp(i\Phi(x,y))\end{gathered} \tag{9}$$

and $A_i(x, y)$ is of finite support, i.e., $A_i(x, y) = 0$ when $\sqrt{x^2 + y^2} > \frac{D_{ap}}{l_{x,y}}$.

$$\begin{gathered}\frac{l_{x,y}^2}{l_z^2}\frac{\partial^2 A}{\partial z^2} + \nabla_\perp^2 A + l_{x,y}^2 k^2 n_0^2 A + 2l_{x,y}^2 k^2 n_0 \sigma_n n(x,y,z)A + l_{x,y}^2 k^2\sigma_n^2 n^2(x,y,z)A = 0 \\ \text{for} \quad -\infty < x, y < \infty, \quad 0 \le z \le \frac{L}{L_0} \quad \text{where} \quad A(x,y,0) = A_i(x,y)\exp(i\Phi(x,y))\end{gathered} \tag{10}$$

and $A_i(x, y)$ is of finite support, i.e., $A_i(x, y) = 0$ when $\sqrt{x^2 + y^2} > \frac{D_{ap}}{l_{x,y}}$.

3.1 Case 1

Letting $l_z = L_0$ and $l_{x,y} = l_0$ gives the following non-dimensional quantities

Symbols	Range
$\epsilon = \frac{l_0}{L_0}$	$10^{-6} - 10^{-3}$
$\alpha = \frac{l_0^2 k}{L_0}$	$10^{-3} - 10$
$\beta = l_0 k$	$10^3 - 10^4$
$\gamma = l_0 k\sqrt{\sigma_n}$	$10^{-1.5} - 10^{1.5}$
$\nu = l_0 k \sigma_n$	$10^{-6} - 10^{-1}$
$l_{pr} = \frac{L}{L_0}$	$10^0 - 10^2$
$d_{ap} = \frac{D_{ap}}{l_0}$	$10^1 - 10^4$

$$\begin{gathered}\epsilon^2\frac{\partial^2 A}{\partial z^2} + \nabla_\perp^2 A + \beta^2 n_0^2 A + 2\gamma^2 n_0 n(x,y,z)A + \nu^2 n^2(x,y,z)A = 0 \\ \text{for} \quad -\infty < x, y < \infty, \quad 0 \le z \le l_{pr} \quad \text{where} \quad A(x,y,0) = A_i(x,y)\exp(i\Phi(x,y))\end{gathered} \tag{11}$$

and $A_i(x, y)$ is of finite support, i.e., $A_i(x, y) = 0$ when $\sqrt{x^2 + y^2} > \frac{D_{ap}}{l_{x,y}}$.

4 Paraxial Approximation

Letting $A = \exp\left(i\frac{\beta}{\epsilon}n_0 z\right)U(x,y,z)$ gives

$$\begin{aligned}
&\epsilon^2\left(-\frac{\beta^2}{\epsilon^2}n_0^2U + 2i\frac{\beta}{\epsilon}n_0\frac{\partial U}{\partial z} + \frac{\partial^2 U}{\partial z^2}\right) + \nabla_\perp^2 U + \beta^2 n_0^2 U + 2\gamma^2 n_0 n(x,y,z)U + \nu^2 n^2(x,y,z)U = 0\\
&2i\beta\epsilon n_0\frac{\partial U}{\partial z} + \epsilon^2\frac{\partial^2 U}{\partial z^2} + \nabla_\perp^2 U + 2\gamma^2 n_0 n(x,y,z)U + \nu^2 n^2(x,y,z)U = 0\\
&2i\alpha n_0\frac{\partial U}{\partial z} + \epsilon^2\frac{\partial^2 U}{\partial z^2} + \nabla_\perp^2 U + 2\gamma^2 n_0 n(x,y,z)U + \nu^2 n^2(x,y,z)U = 0\\
&\quad \text{for} \quad -\infty < x,y < \infty, \quad 0 \le z \le l_{pr} \quad \text{where} \quad U(x,y,0) = U_i(x,y)\exp(i\Phi(x,y))
\end{aligned} \tag{12}$$

and $U_i(x,y)$ is of finite support, i.e., $U_i(x,y) = 0$ when $\sqrt{x^2+y^2} > d_{ap}$.

Taking the dominate balance to be $\{\nu^2 \sim \epsilon^2, \alpha \sim \gamma^2 \sim 1\}$ gives at first order

$$\boxed{\begin{aligned}
&2i\alpha n_0\frac{\partial U}{\partial z} + \nabla_\perp^2 U + 2\gamma^2 n_0 n(x,y,z)U = 0 && (13)\\
&\quad \text{for} \quad -\infty < x,y < \infty, \quad 0 \le z \le l_{pr} \quad \text{where} \quad U(x,y,0) = U_i(x,y)\exp(i\Phi(x,y)) && (14)
\end{aligned}}$$

4.1 Nondimensionalized Correlation

$$C_{cor}(\|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\|) = 1 - \frac{C_n^2}{2\sigma_n^2}\begin{cases} l_0^{-4/3}\|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\|^2 & 0 < \|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\| \le l_0 \\ \|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\|^{2/3} & l_0 \le \|\tilde{\boldsymbol{r}}_1 - \tilde{\boldsymbol{r}}_2\| \le L_0 \end{cases} \tag{15}$$

$$C_{cor}(\|\boldsymbol{r}_1 - \boldsymbol{r}_2\|) = 1 - \frac{C_n^2}{2\sigma_n^2}\begin{cases} \frac{L_0^2}{l_0^{4/3}}\left((z_1-z_2)^2 + \epsilon^2\left|\boldsymbol{r}_{\perp,1} - \boldsymbol{r}_{\perp,2}\right|^2\right) & 0 \le \sqrt{(z_1-z_2)^2 + \epsilon^2\left|\boldsymbol{r}_{\perp,1} - \boldsymbol{r}_{\perp,2}\right|^2} \le \epsilon \\ L_0^{2/3}\left((z_1-z_2)^2 + \epsilon^2\left|\boldsymbol{r}_{\perp,1} - \boldsymbol{r}_{\perp,2}\right|^2\right)^{1/3} & \epsilon \le \sqrt{(z_1-z_2)^2 + \epsilon^2\left|\boldsymbol{r}_{\perp,1} - \boldsymbol{r}_{\perp,2}\right|^2} \le 1 \end{cases} \tag{16}$$

$$C_{cor}(\|\boldsymbol{r}_1 - \boldsymbol{r}_2\|) = 1 - \frac{1}{C_0^{2/3}}\begin{cases} \frac{L_0^{4/3}}{l_0^{4/3}}\left((z_1-z_2)^2 + \epsilon^2\left|\boldsymbol{r}_{\perp,1} - \boldsymbol{r}_{\perp,2}\right|^2\right) & 0 \le \sqrt{(z_1-z_2)^2 + \epsilon^2\left|\boldsymbol{r}_{\perp,1} - \boldsymbol{r}_{\perp,2}\right|^2} \le \epsilon \\ \left((z_1-z_2)^2 + \epsilon^2\left|\boldsymbol{r}_{\perp,1} - \boldsymbol{r}_{\perp,2}\right|^2\right)^{1/3} & \epsilon \le \sqrt{(z_1-z_2)^2 + \epsilon^2\left|\boldsymbol{r}_{\perp,1} - \boldsymbol{r}_{\perp,2}\right|^2} \le 1 \end{cases} \tag{17}$$

4.1.1 Delta Correlation in z

$$C_{cor}(x_1 - x_2, y_1 - y_2) = 1 - \frac{1}{C_0^{2/3}}\begin{cases} \frac{L_0^{4/3}}{l_0^{4/3}}\epsilon^2\left((x_2-x_1)^2 + (y_2-y_2)^2\right) & 0 \le \sqrt{((x_2-x_1)^2 + (y_2-y_2)^2)} \le 1 \\ \epsilon^{2/3}\left((x_2-x_1)^2 + (y_2-y_2)^2\right)^{1/3} & 1 \le \sqrt{((x_2-x_1)^2 + (y_2-y_2)^2)} \le \frac{1}{\epsilon} \end{cases} \tag{18}$$

$$C_{cor}(x_1 - x_2, y_1 - y_2) = 1 - \frac{\epsilon^{2/3}}{C_0^{2/3}}\begin{cases} \left((x_2-x_1)^2 + (y_2-y_2)^2\right) & 0 \le \sqrt{((x_2-x_1)^2 + (y_2-y_2)^2)} \le 1 \\ \left((x_2-x_1)^2 + (y_2-y_2)^2\right)^{1/3} & 1 \le \sqrt{((x_2-x_1)^2 + (y_2-y_2)^2)} \le \frac{1}{\epsilon} \end{cases} \tag{19}$$

5 Pseudo-spectral Method

$$\begin{gathered}\frac{\partial U}{\partial z} = i\frac{1}{2n_0\alpha}\nabla_\perp^2 U + i\frac{\gamma^2}{\alpha}n(x,y,z)U \\ \text{for} \quad -\infty < x, y < \infty, \quad 0 < z < \frac{L}{L_0} = l_{pr} \quad \text{where} \quad U(x,y,0) = U_i(x,y)\exp(i\Phi(x,y))\end{gathered} \tag{20}$$

and $U_i(x,y)$ is of finite support, i.e., $U_i(x,y) = 0$ when $\sqrt{x^2+y^2} > \frac{D_i}{l_0} = \tilde{D}_i$.

Take the FT of 20 in x, y:

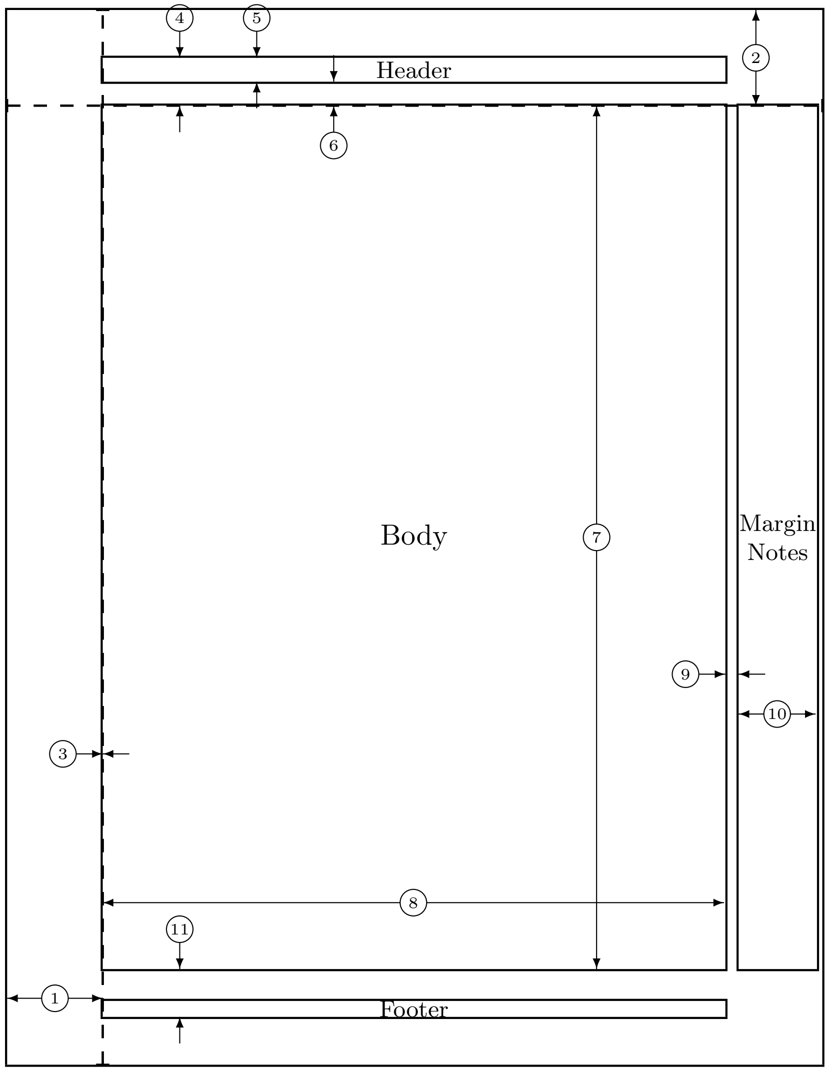
$$\begin{gathered}\frac{\partial \hat{U}}{\partial z} = -i\frac{1}{2n_0\alpha}(k_x^2+k_y^2)\hat{U} + i\frac{\gamma^2}{\alpha}\mathcal{F}\Big[n(x,y,z)\mathcal{F}^{-1}\Big[\hat{U}\Big]\Big] \\ \text{for} \quad 0 < z < \frac{L}{L_0} = l_{pr} \quad \text{where} \quad \hat{U}(k_x,k_y,0) = \mathcal{F}[U_i(x,y)\exp(i\Phi(x,y))]\end{gathered} \tag{21}$$

then let $\hat{U} = \exp\Big(-i\frac{1}{2n_0\alpha}(k_x^2+k_y^2)z\Big)\hat{V}$ giving

$$\begin{gathered}\frac{\partial \hat{V}}{\partial z} = i\frac{\gamma^2}{\alpha}\exp\left(i\frac{1}{2n_0\alpha}(k_x^2+k_y^2)z\right)\mathcal{F}\left[n(x,y,z)\mathcal{F}^{-1}\left[\exp\left(-i\frac{1}{2n_0\alpha}(k_x^2+k_y^2)z\right)\hat{V}\right]\right] \\ \text{for} \quad 0 < z < \frac{L}{L_0} = l_{pr} \quad \text{where} \quad \hat{U}(k_x,k_y,0) = \mathcal{F}[U_i(x,y)\exp(i\Phi(x,y))]\end{gathered} \tag{22}$$

References

[1] J. W. Goodman. *Statistical optics*. 1985.



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7	\textheight = 650pt	8	\textwidth = 469pt
9	\marginparsep = 10pt	10	\marginparwidth = 59pt
11	\footskip = 36pt		\marginparpush = 5pt (not shown)
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